

Feature Engineering and Feature Selection Validation Report

Model Name

Mortgage Insurance Credit Risk Model

Objective

The purpose of this review is to assess whether the feature engineering and feature selection process is methodologically sound, business-justified, reproducible, and free from leakage, bias, or unnecessary model instability.

The review focused on:

- Engineered feature construction
- Variable transformations
- Missing-value handling
- Outlier treatment
- Ratio and interaction variables
- Economic feature generation
- Borrower aggregation features
- Leakage assessment
- Feature stability and interpretability

The assessment was performed using the implemented preprocessing and feature engineering framework.

1. Feature Creation and Transformation

Validation Assessment

Review Item	Status
All engineered features are documented	Partial
Business rationale for engineered features is reasonable	Yes
Transformations preserve economic meaning	Yes

Review Item	Status
Ratios and derived variables are appropriate	Yes
Derived features align with intended use	Yes
No material distortion of risk relationship identified	Yes

Validator Comments

The framework creates several engineered features relevant to mortgage credit risk, including:

- Loan-to-income ratio (LTI)
- Monthly interest rate
- Monthly payment calculation
- Property age
- Property value
- Rural classification
- Economic region assignment

The engineered variables are economically meaningful and generally consistent with mortgage underwriting practices.

Some feature definitions and business rationales should be documented more clearly for governance purposes.

2. Missing-Value Related Features

Validation Assessment

Review Item	Status
Missing-value flags considered where appropriate	Partial
Missingness assessed as potentially informative	Partial
Imputation logic documented and justified	Partial

Review Item	Status
Imputation avoids future information	Yes
Missing-value treatment consistent across datasets	Yes

Validator Comments

The framework uses several fallback methods to reduce missing values, including:

- Bureau score substitution
- Group-level propagation
- Formula-based reconstruction
- Group-average replacement

Credit scores are derived from multiple bureau sources using fallback logic. Missing property values are reconstructed using LTV and insurance information.

No evidence of future information usage was identified in the imputation logic.

3. Outlier-Related Features and Handling

Validation Assessment

Review Item	Status
Extreme values reviewed for business validity	Yes
Capping and truncation documented	Partial
Tail-risk observations preserved appropriately	Partial
Outlier indicator variables justified	Partial
Outlier treatment consistent across datasets	Yes

Validator Comments

The framework applies range clipping and hard limits for:

- Credit scores
- Property values

- Contract rates
- Loan terms

Credit scores outside acceptable ranges are replaced with missing values. Interest rates are normalized to realistic values.

The validator considers the outlier handling reasonable; however, selected thresholds should be formally documented.

4. Categorical Variable Handling

Validation Assessment

Review Item	Status
Category grouping documented	Partial
Rare categories handled appropriately	Partial
Ordinal and nominal variables treated appropriately	Yes
Encoding method suitable for model type	Yes
Encoding avoids leakage	Yes
Category definitions stable over time	Partial

Validator Comments

The framework standardizes several categorical variables, including:

- Loan purpose
- Portability indicators
- Occupancy indicators
- Tenure mapping
- Regional assignment

Unknown categories are replaced using standardized labels. Regional segmentation and rural classification are applied consistently.

Additional documentation of category grouping logic would improve transparency.

5. Time-Based Feature Engineering

Validation Assessment

Review Item	Status
Date variables converted appropriately	Yes
Lag features use prior information only	Partial
Rolling features constructed appropriately	Partial
Vintage effects considered	Yes
No future information used in time-based features	Partial

Validator Comments

The framework creates time-based features using:

- RECEIVED_SID
- BOOKYEAR
- Economic lag variables
- Economic forward-shift variables

Economic variables include lagged and shifted HPI and unemployment measures.

Potential leakage risk exists if forward-looking economic variables are not aligned properly with the application date. Additional validation is recommended.

6. Interaction and Ratio Features

Validation Assessment

Review Item	Status
Interaction terms are meaningful	Yes
Ratios are mathematically correct	Yes

Review Item	Status
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Multicollinearity assessed where applicable	Partial
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Interaction features improve interpretability	Yes
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Number of interaction features reasonable	Yes
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Validator Comments

The framework creates several economically meaningful ratio features, including:

- Loan-to-income ratio
- Monthly payment calculations
- Property value reconstruction
- Weighted bureau score combinations

The engineered features are aligned with mortgage underwriting and borrower affordability assessment.

No excessive feature complexity was identified.

7. Binning and Discretization

Validation Assessment

Review Item	Status
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Bin definitions documented	Partial
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Bins based on business or statistical logic	Yes
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Bins are interpretable	Yes
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Bin boundaries reasonable	Yes
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Binning consistent across datasets	Yes
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Validator Comments

The framework primarily relies on normalization and clipping rather than aggressive binning. Certain variables are grouped into standardized business categories such as:

- Rural vs urban
- Tenure type
- Loan purpose

The implemented grouping logic appears reasonable and interpretable.

8. Feature Leakage Assessment

Validation Assessment

Review Item	Status
Candidate features reviewed for leakage	Partial
No post-outcome information identified	Yes
No proxy outcome variables identified	Partial
Feature construction respects as-of date	Partial
Leakage checks performed on engineered features	Partial

Validator Comments

No obvious target leakage was identified within the preprocessing and feature engineering logic.

However, the framework includes forward-looking economic shift variables that may introduce leakage risk if not aligned correctly with application timing.

Additional formal leakage testing is recommended.

9. Feature Redundancy and Correlation

Validation Assessment

Review Item	Status
Highly correlated variables identified	Partial
Redundant features reviewed	Partial

Review Item	Status
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Correlation impact assessed	Partial
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Overlapping information rationalized	Partial
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Feature complexity appropriate	Yes
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Validator Comments

The framework combines multiple bureau variables and borrower-level statistics. While the resulting feature set appears manageable, no formal correlation analysis or redundancy review was identified in the reviewed implementation.

Additional feature correlation analysis is recommended before final model development.

10. Feature Selection Methodology

Validation Assessment

Review Item	Status
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Feature selection criteria documented	Partial
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Selection method appropriate	Yes
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Domain knowledge considered	Yes
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Statistical selection methods applied consistently	Partial
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Final feature set defensible	Yes
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Excluded features documented	No
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Validator Comments

The feature set appears to be driven primarily by mortgage underwriting knowledge and business relevance.

The framework includes:

- Credit bureau variables
- Affordability measures

- Property characteristics
- Regional economic variables
- Borrower aggregation features

The validator considers the feature set generally appropriate for mortgage insurance credit risk modeling. However, formal documentation of feature selection criteria and excluded variables would strengthen governance.

11. Stability and Robustness of Features

Validation Assessment

Review Item	Status
Feature distributions assessed over time	Partial
Stability reviewed by segment and vintage	Partial
Features stable across sample periods	Partial
Features robust to small data changes	Yes
Stability concerns documented	Partial

Validator Comments

The framework includes multiple years of data and economic-region segmentation, which supports stability assessment.

However, no formal feature stability monitoring or population stability testing was identified during review. Additional monitoring procedures are recommended.

12. Interpretability and Governance

Validation Assessment

Review Item	Status
Final feature set explainable to stakeholders	Yes
Features comply with model risk standards	Partial

Review Item	Status
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Features appropriate for model materiality	Yes
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Expert overrides documented	Partial
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Features support monitoring and revalidation	Yes
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Validator Comments

Most engineered features are economically interpretable and aligned with mortgage underwriting practices. Examples include:

- Borrower affordability ratios
- Property characteristics
- Regional economic indicators
- Credit bureau metrics

The overall feature set appears appropriate for governance review and ongoing model monitoring.

13. Validation Conclusion

Validation Assessment

Conclusion Item	Status
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Feature engineering is acceptable	Yes
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Feature engineering acceptable subject to remediation	Yes
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Feature engineering is not acceptable	No
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Conclusion

The feature engineering and feature selection framework is generally appropriate for mortgage insurance credit risk modeling.